Global Tech Skill Trends

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OUTLINE

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- Methodology
- Results
  - Visualization – Charts
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- Discussion
  - Findings & Implications
- Conclusion
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EXECUTIVE SUMMARY

- Goal: stay competitive in the rapidly evolving technology landscape
- Objective: identify emerging and future tech skills
- Data Sources:
  - Job Postings
  - Training Portals
  - Stack Overflow Developer Survey
- Key Findings:
  - JavaScript: most trending programming language
  - PostgreSQL: most trending Database
  - AWS: most trending Platform

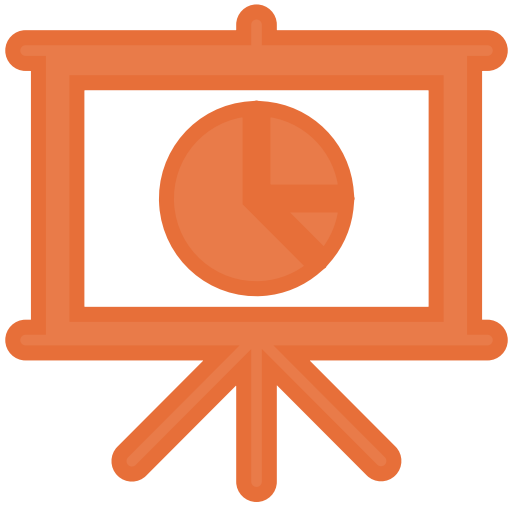
INTRODUCTION

- Stack Overflow's annual Developer Survey is the largest and most comprehensive survey of people who code around the world.
- Over 60,000 developers
- Identify Trends in:
  - Technology:
    - Programming language
    - Database
    - Platform
    - Webframe
  - Work Preferences
  - Job Satisfaction
  - Compensation

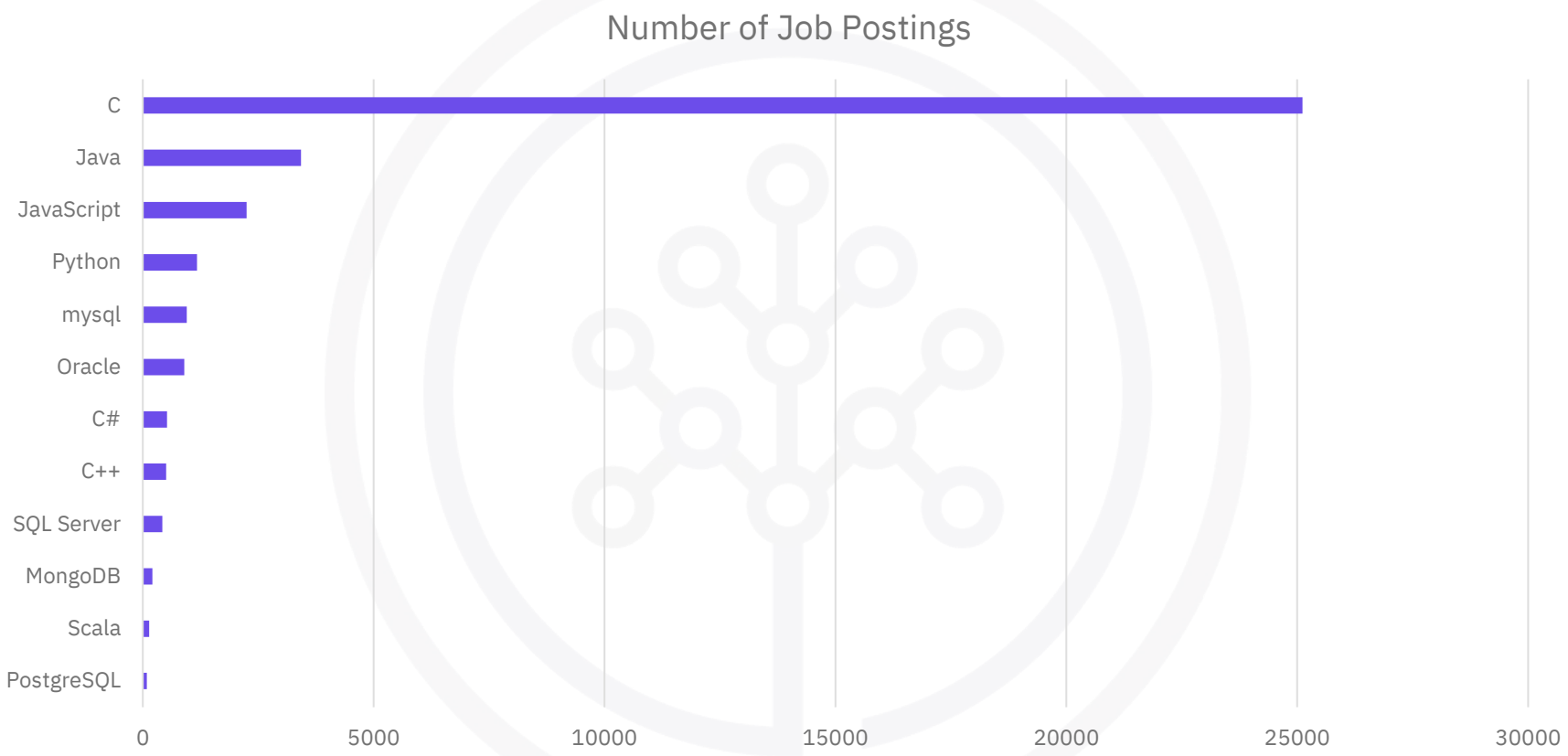
METHODOLOGY

- Data Sources:
  - Job Postings
  - Training Portals
  - Stack Overflow Developer Survey
- Data Collection:
  - Web Scraping
  - Web APIs
- Data Wrangling: handling outliers, duplicates, ...
- Exploratory Data Analysis:
  - Analyzing data distribution
  - Analyzing correlations
- Data Visualization: charts, graphs and dashboards

Results

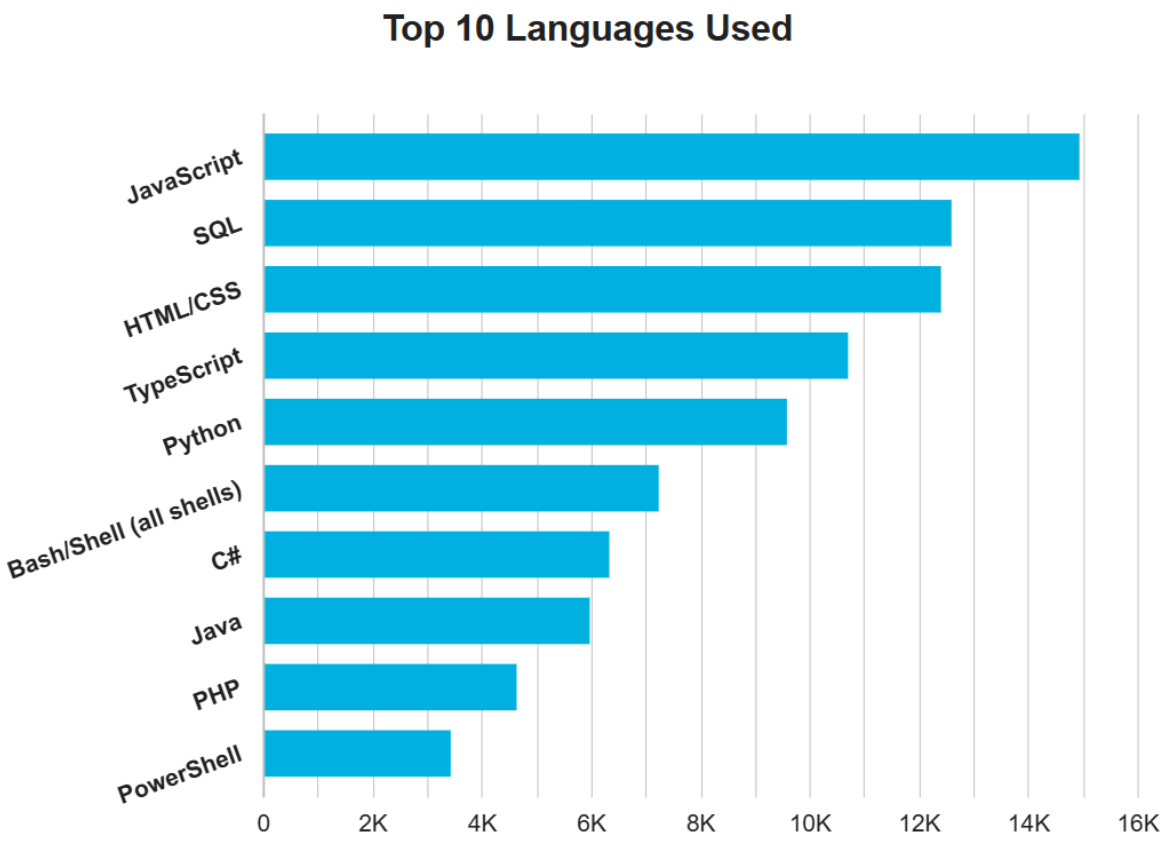


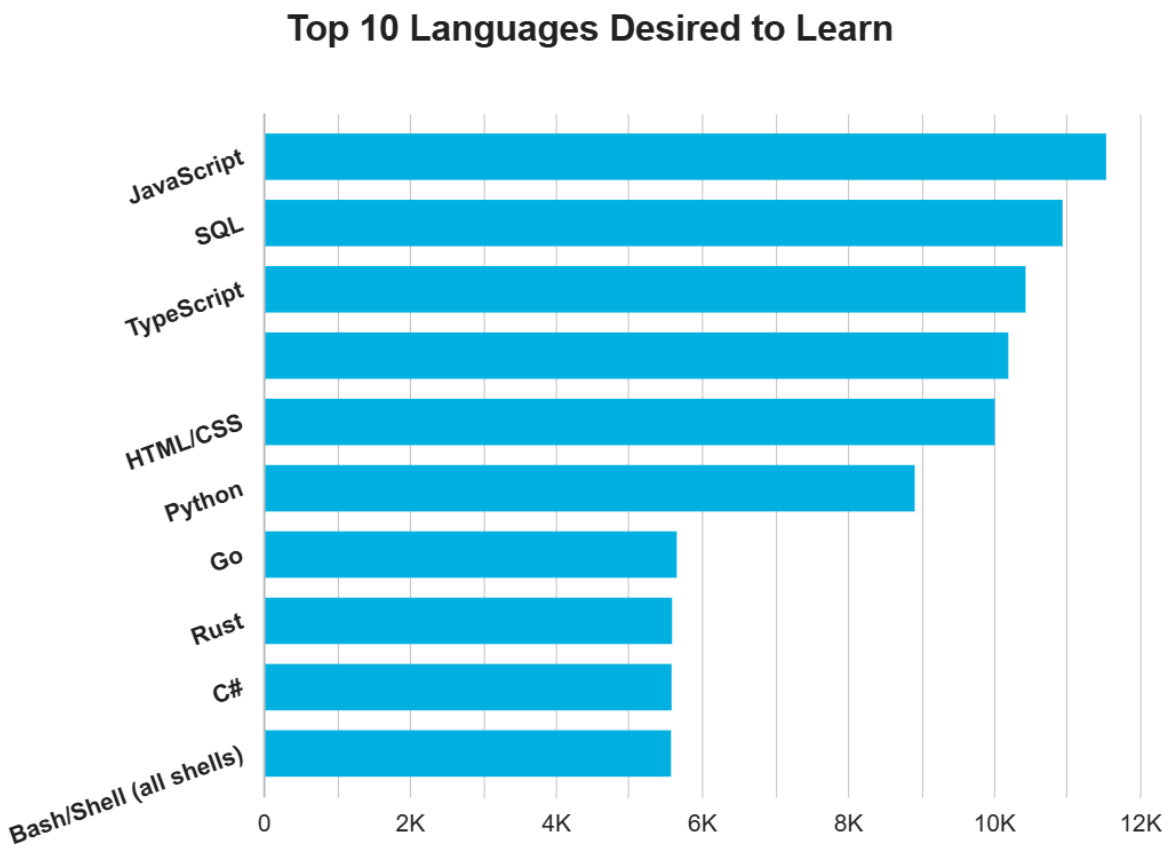
JOB POSTINGS



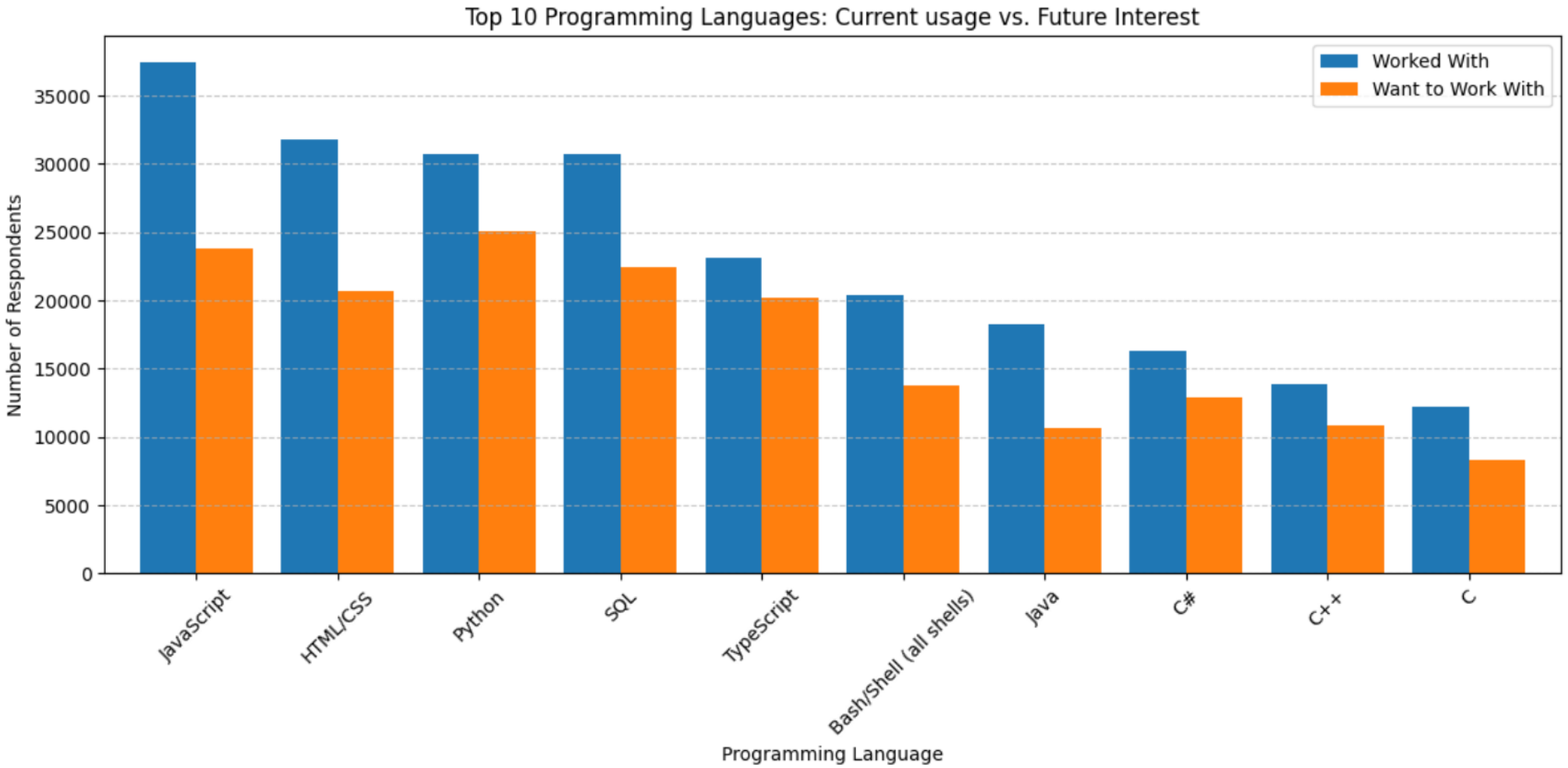
POPULAR LANGUAGES

PROGRAMMING LANGUAGE TRENDS





PROGRAMMING LANGUAGE TRENDS



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

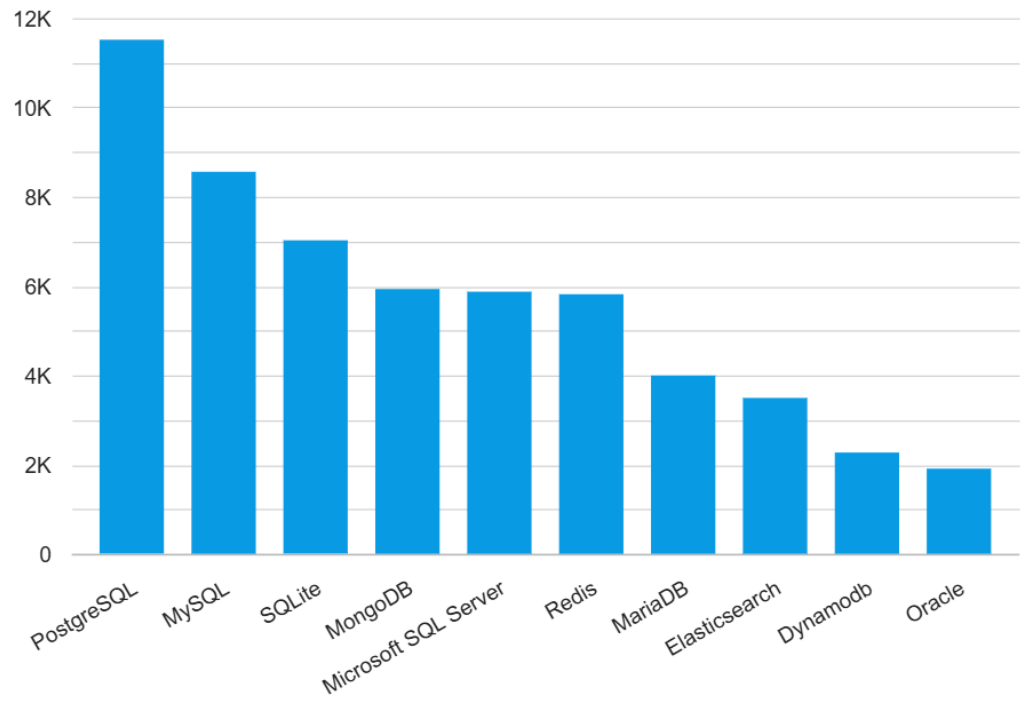
Findings

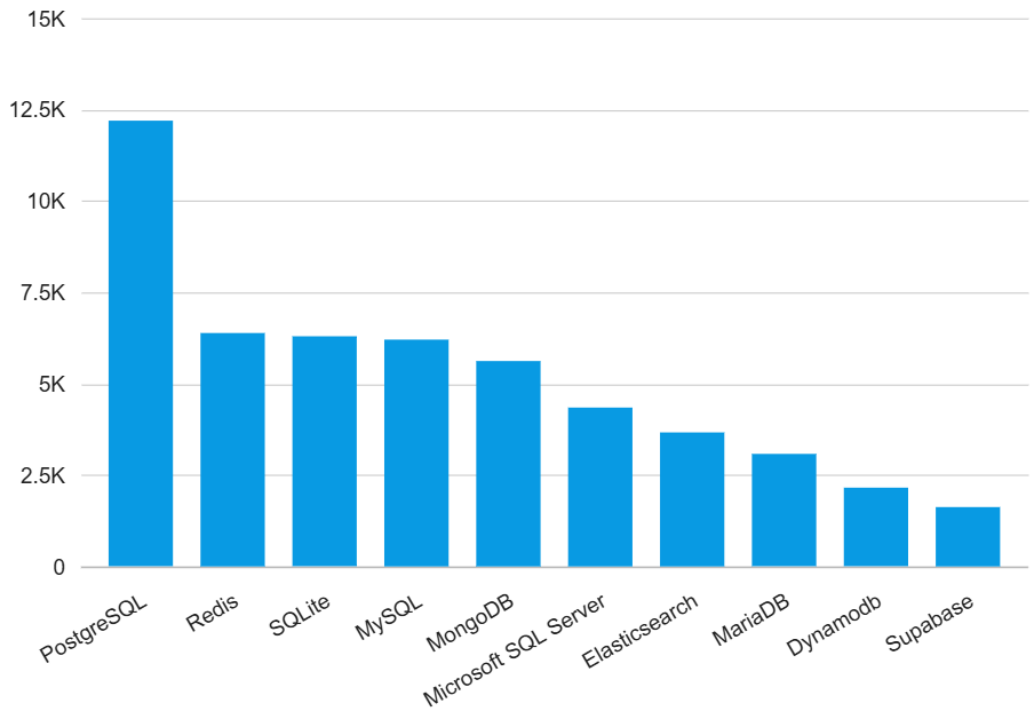
- JavaScript is the most popular language and is going to maintain its place
- Other scripting languages like HTML/CSS, SQL and TypeScript are also very popular
- Python is the most popular among non-scripting languages

Implications

- Go and Rust are gaining momentum in popularity
- PHP and Shell(s) are losing their places
- Web-related technologies are the top-rated skills

DATABASE TRENDS





DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is the most popular DB and is going to maintain its place
- MySQL and SQLite are the next-hot DBs
- Oracle is loosing its old-day popularity

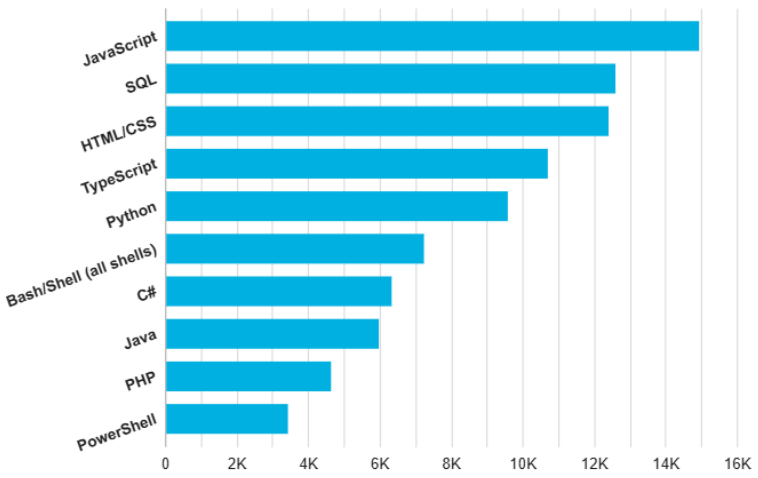
Implications

- Redis is expected to be one of the hottest DBs
- Supabase is gaining popularity

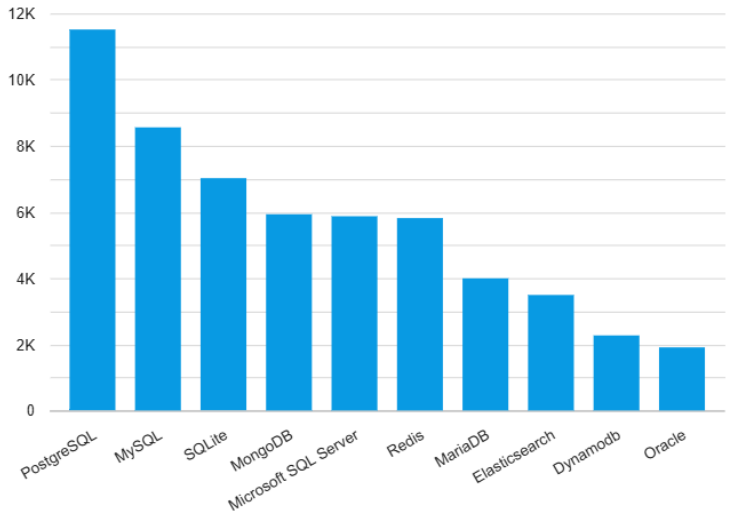
DASHBOARD

DASHBOARD TAB 1: Current Technology Usage

Top 10 Languages Used

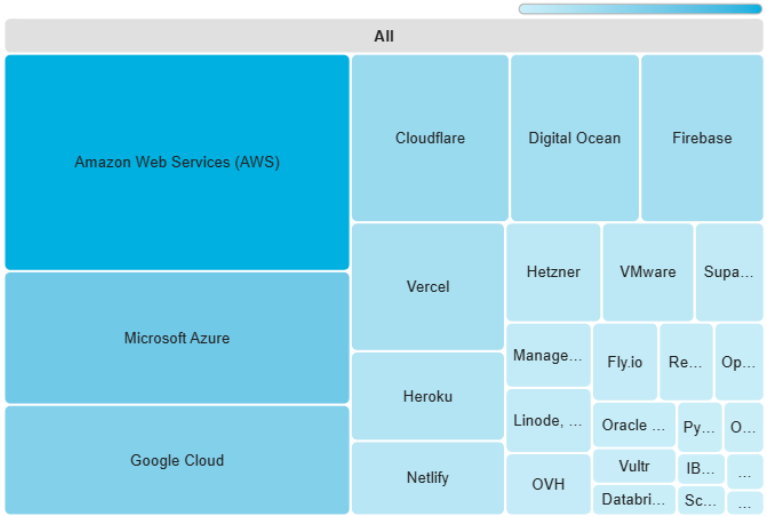


Top 10 Databases Used



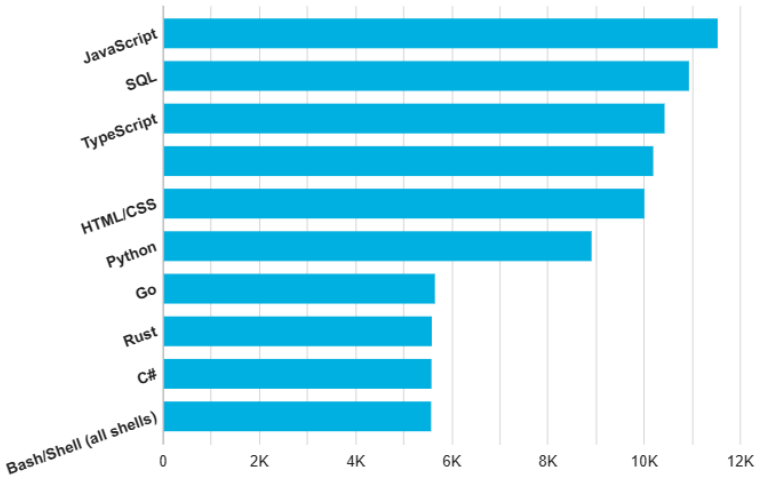
Webframe Used

Platform Used

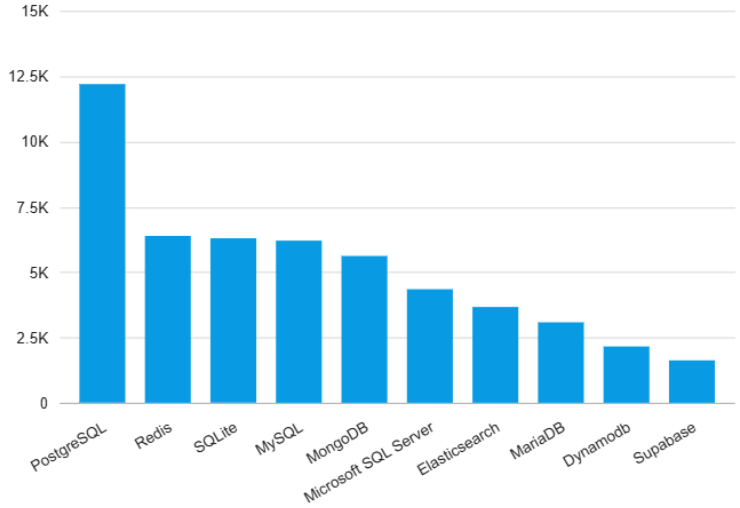


DASHBOARD TAB 2: Future Technology Trends

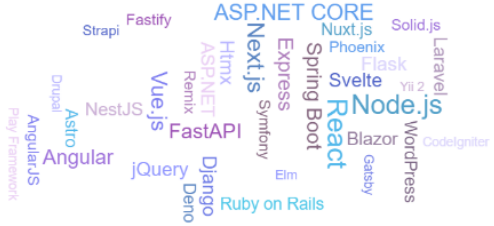
Top 10 Languages Desired to Learn



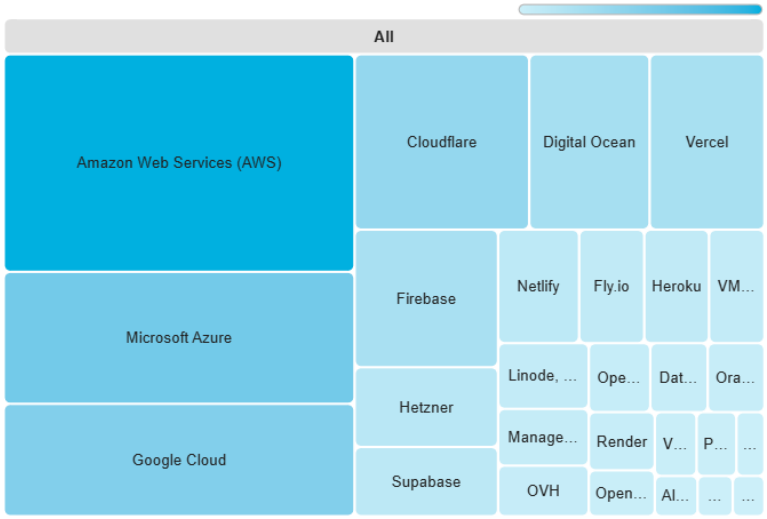
Top 10 Databases Desired to Learn



Webframe Desired to Learn

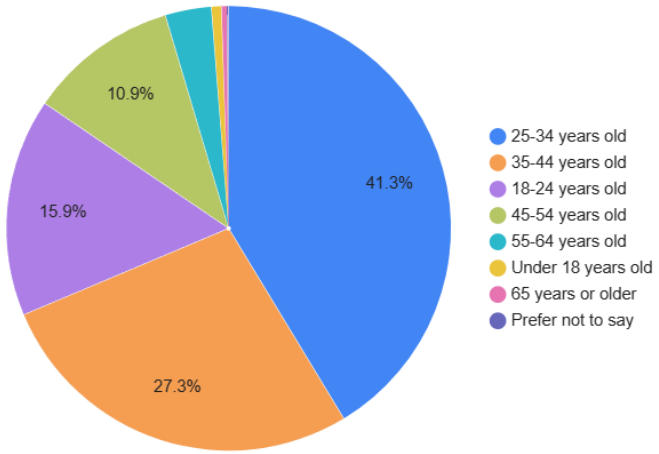


Platform Desired to Learn

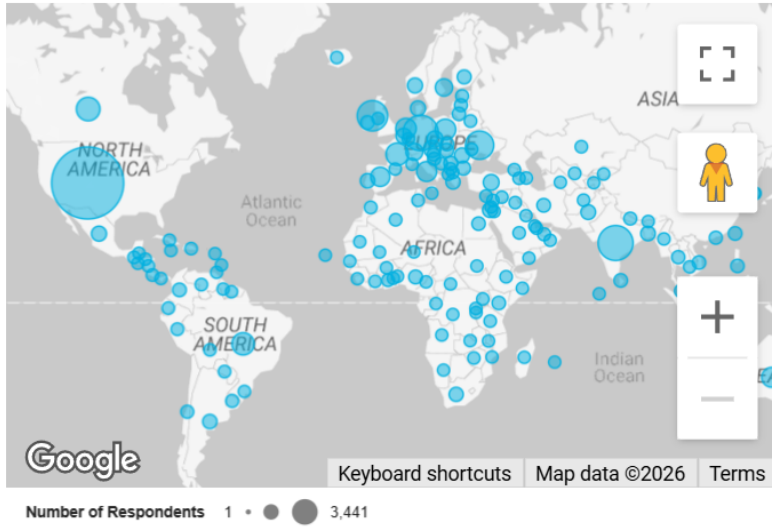


DASHBOARD TAB 3: Demographics

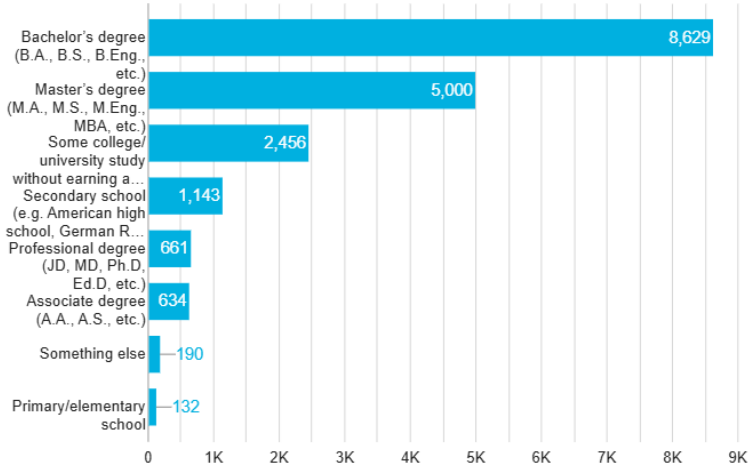
Respondents by Age



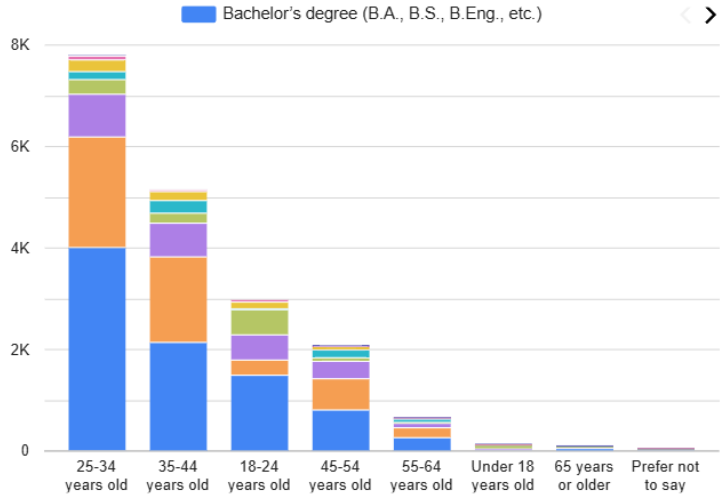
Respondent by Country



Respondent Distribution by Education Level



Respondent by Age and Education Level



DISCUSSION

OVERALL FINDINGS & IMPLICATIONS

Findings

- AWS, Azure and Google Cloud are the most popular platforms
- React and Node.js are the most popular webframes
- C and C# are still highly demanded skills

Demographics

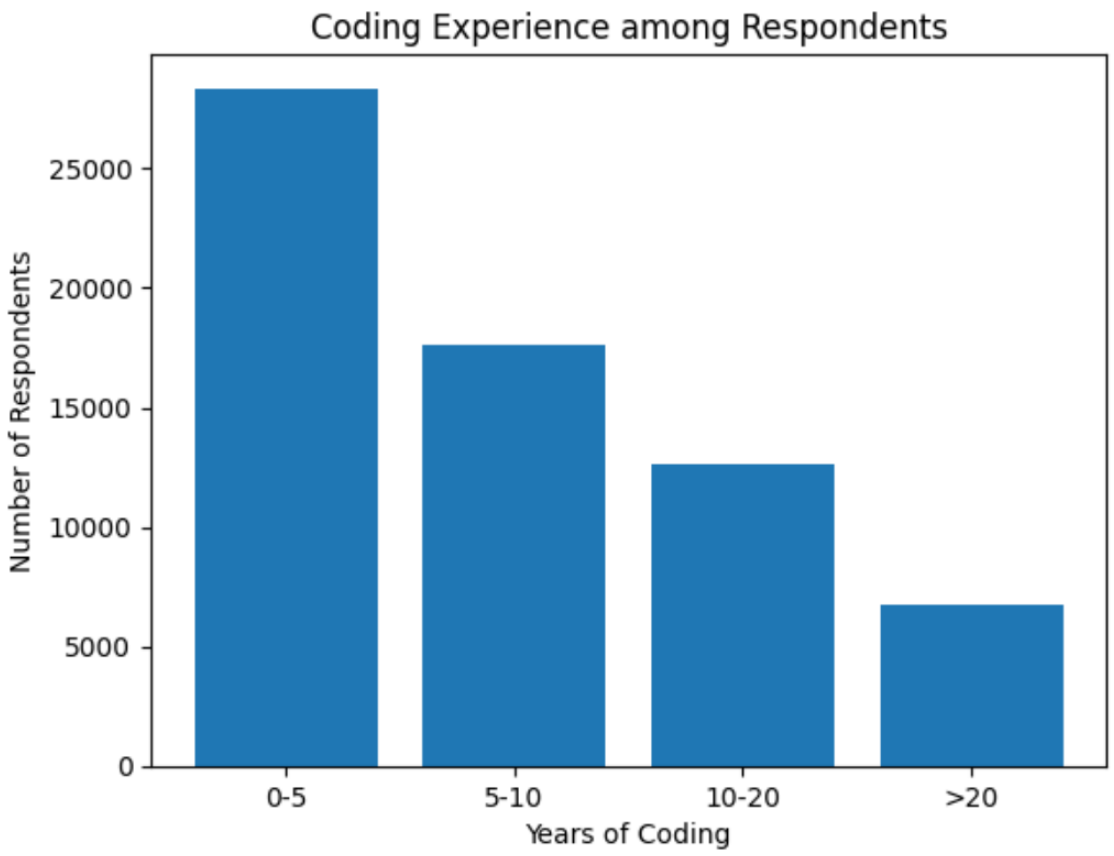
- Most survey participants (~70%) are 25-44 Years Old
- 90% of respondents have university degrees
- Programmers are mostly lived in the US, Europe and India

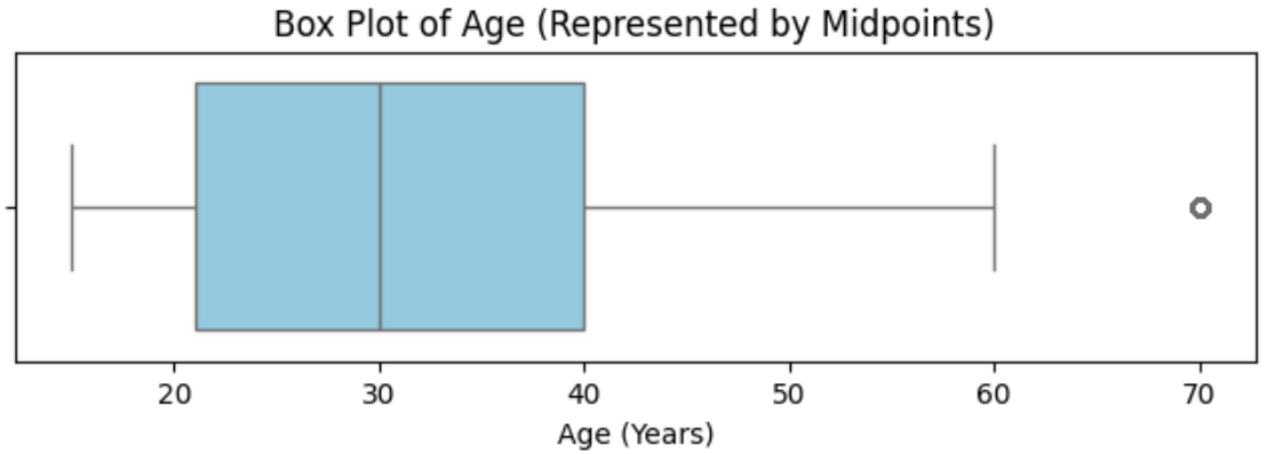
CONCLUSION

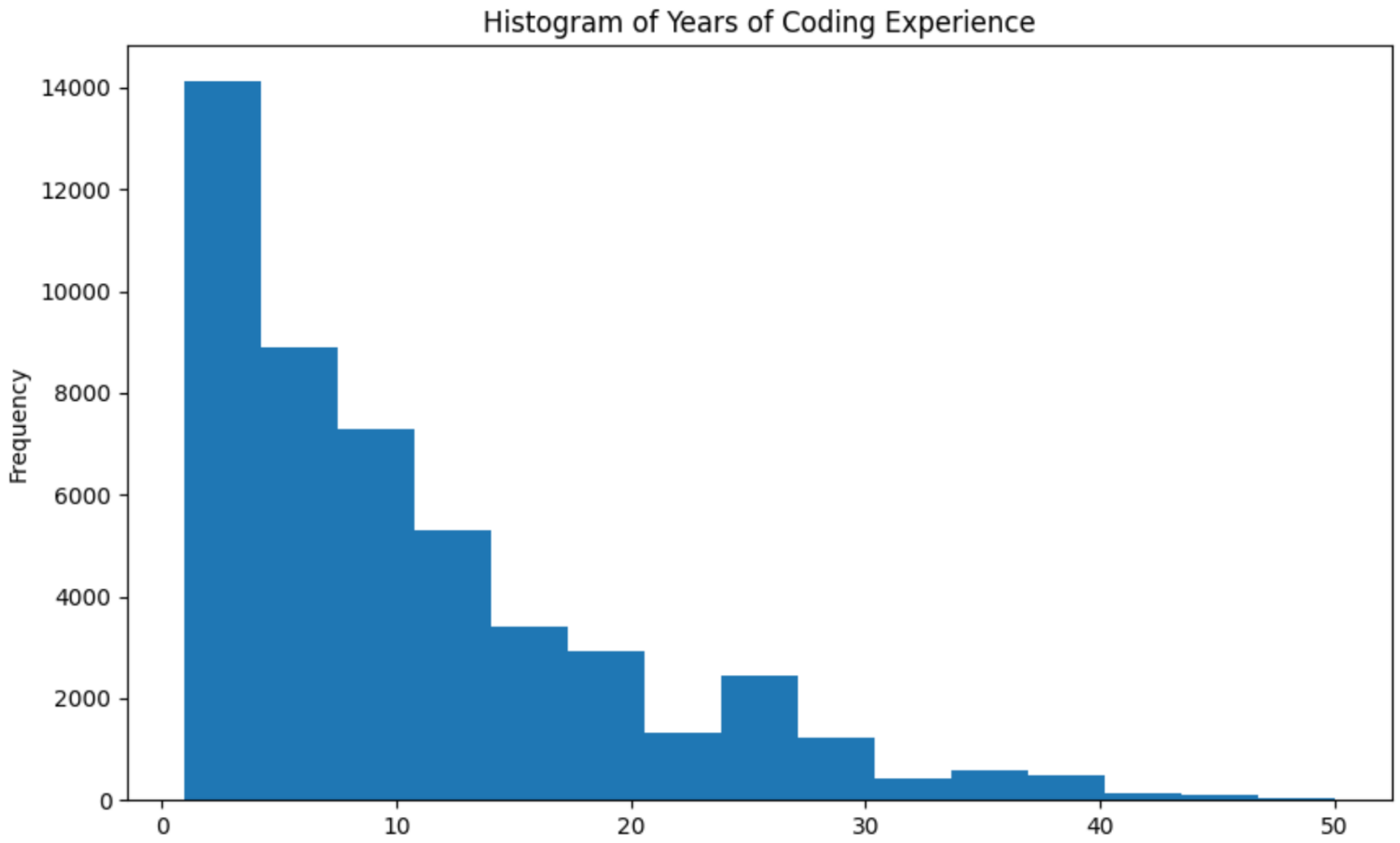
Most Trending Technologies:

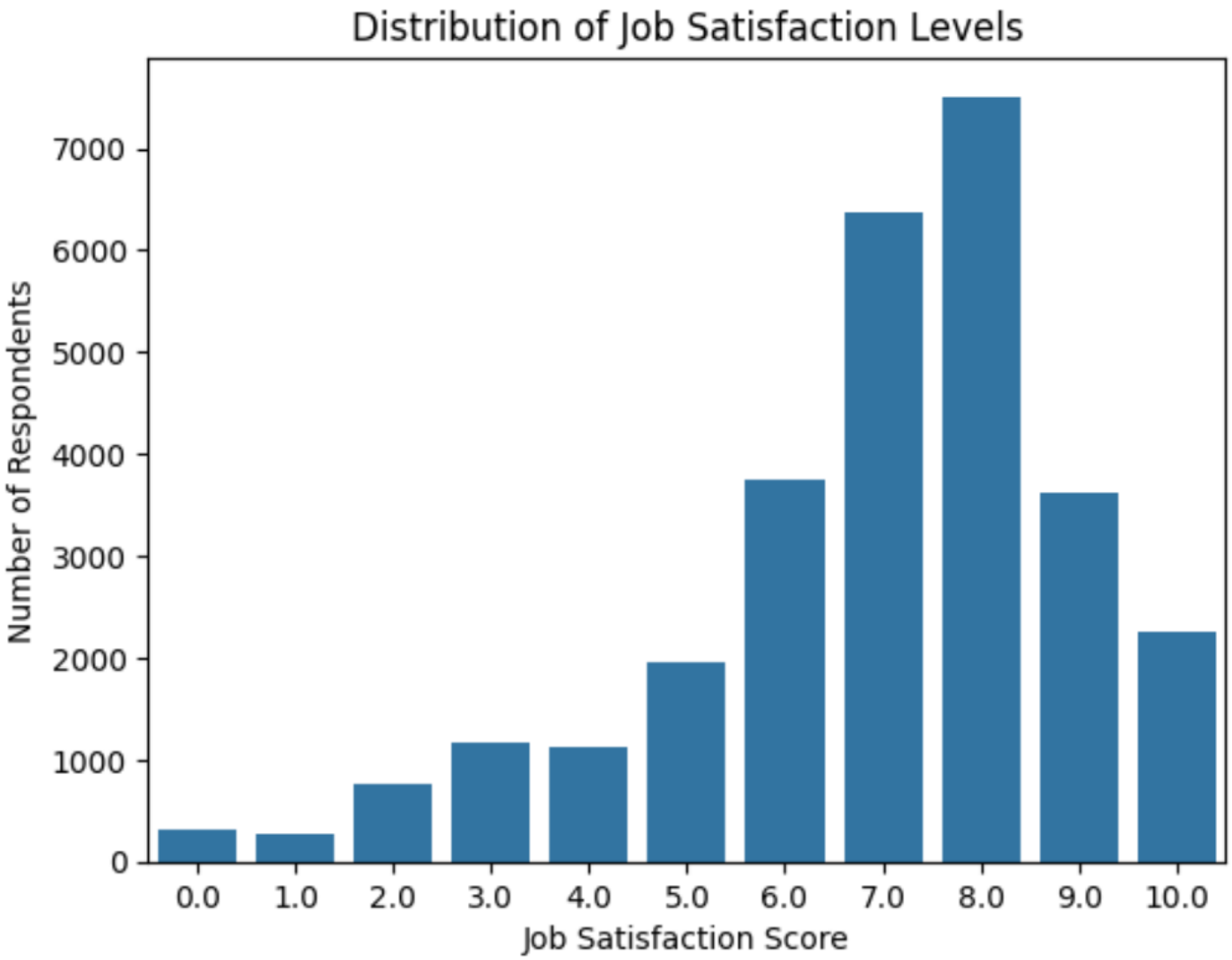
- JavaScript, HTML, Python
- PostgreSQL, MySQL
- AWS, Azure
- React , Node.js

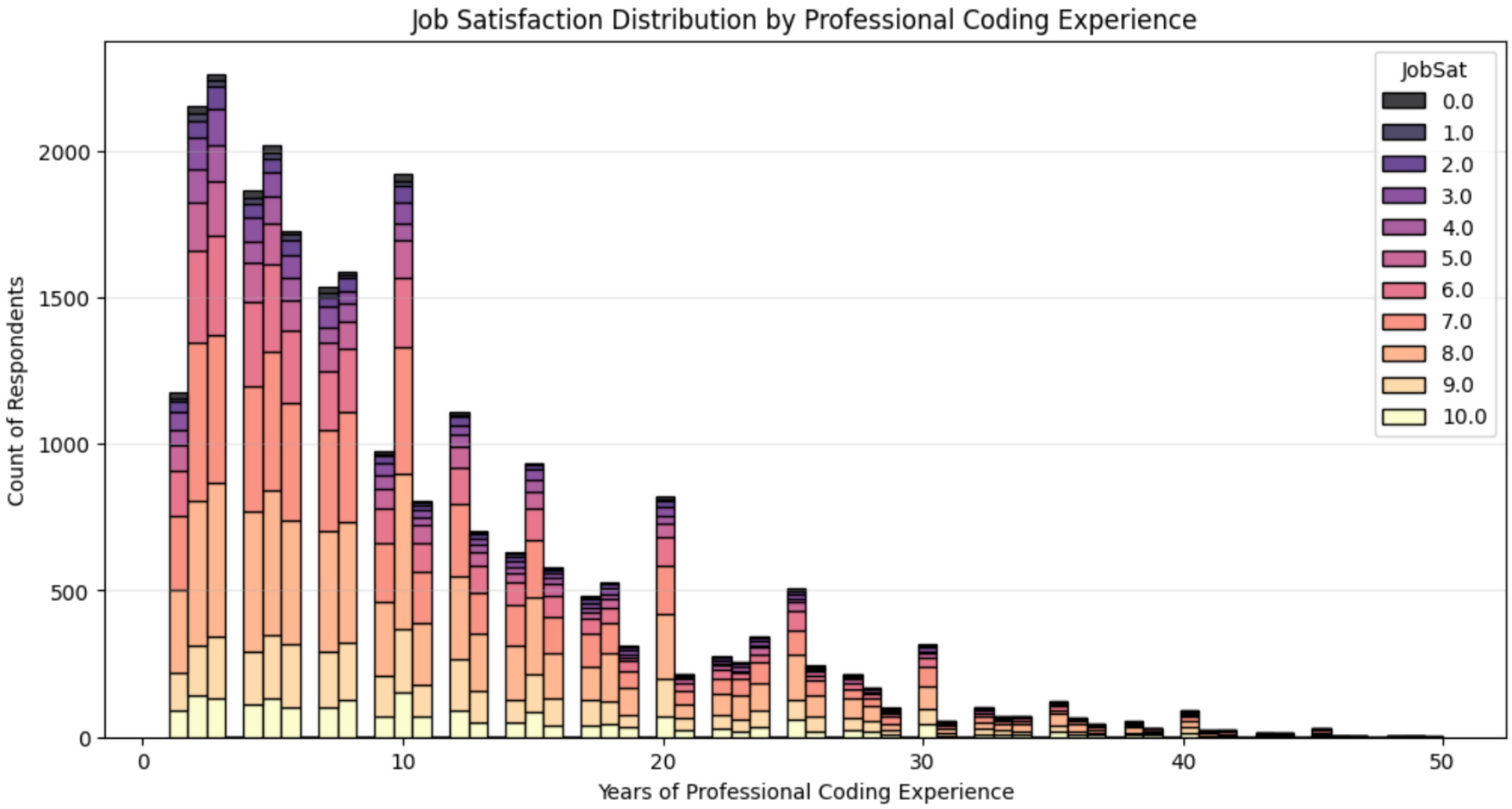
APPENDIX

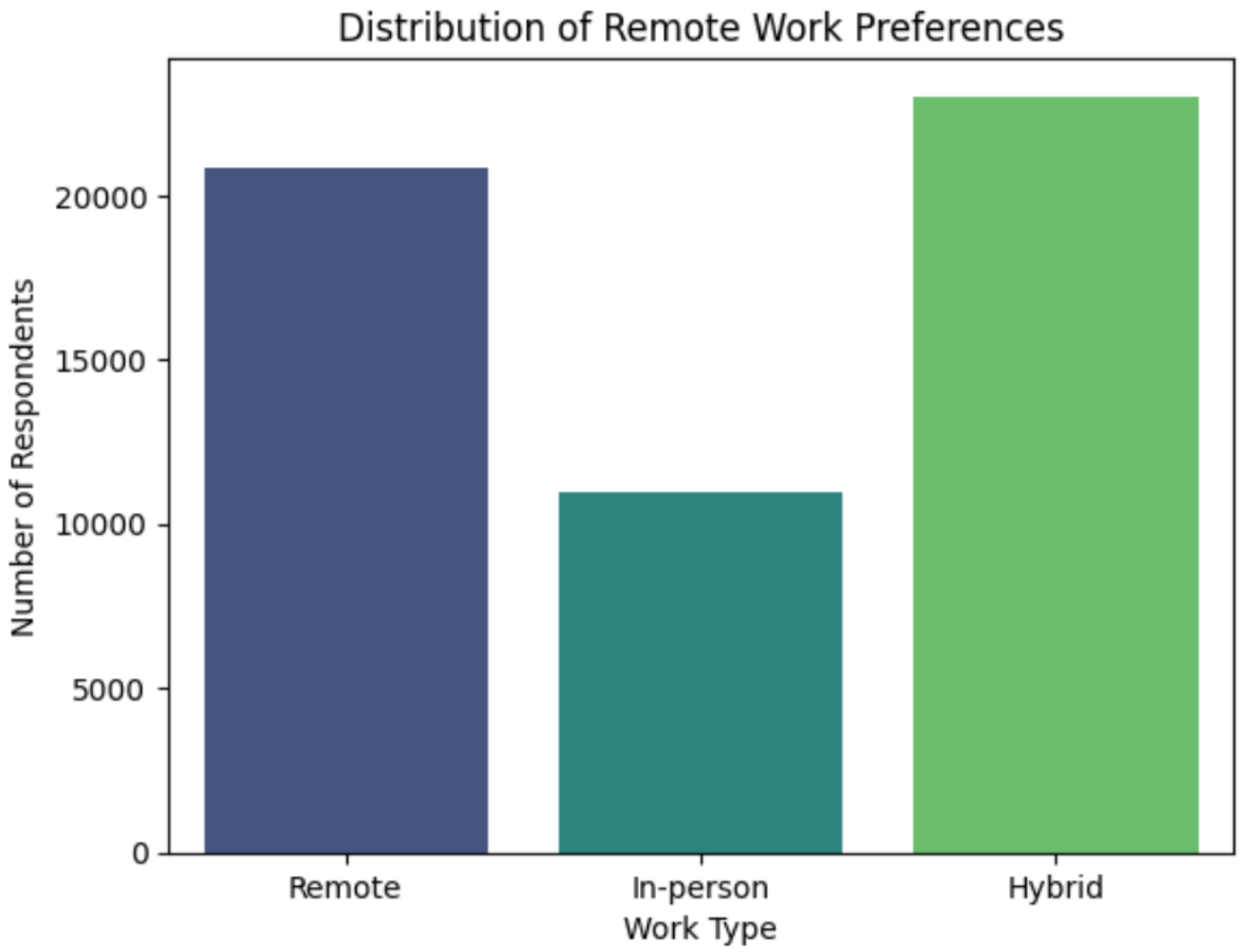

Coding Experience among Respondents
Number of Respondents
25000
20000
15000
10000
5000
0
0-5
5-10
10-20
>20
Years of Coding

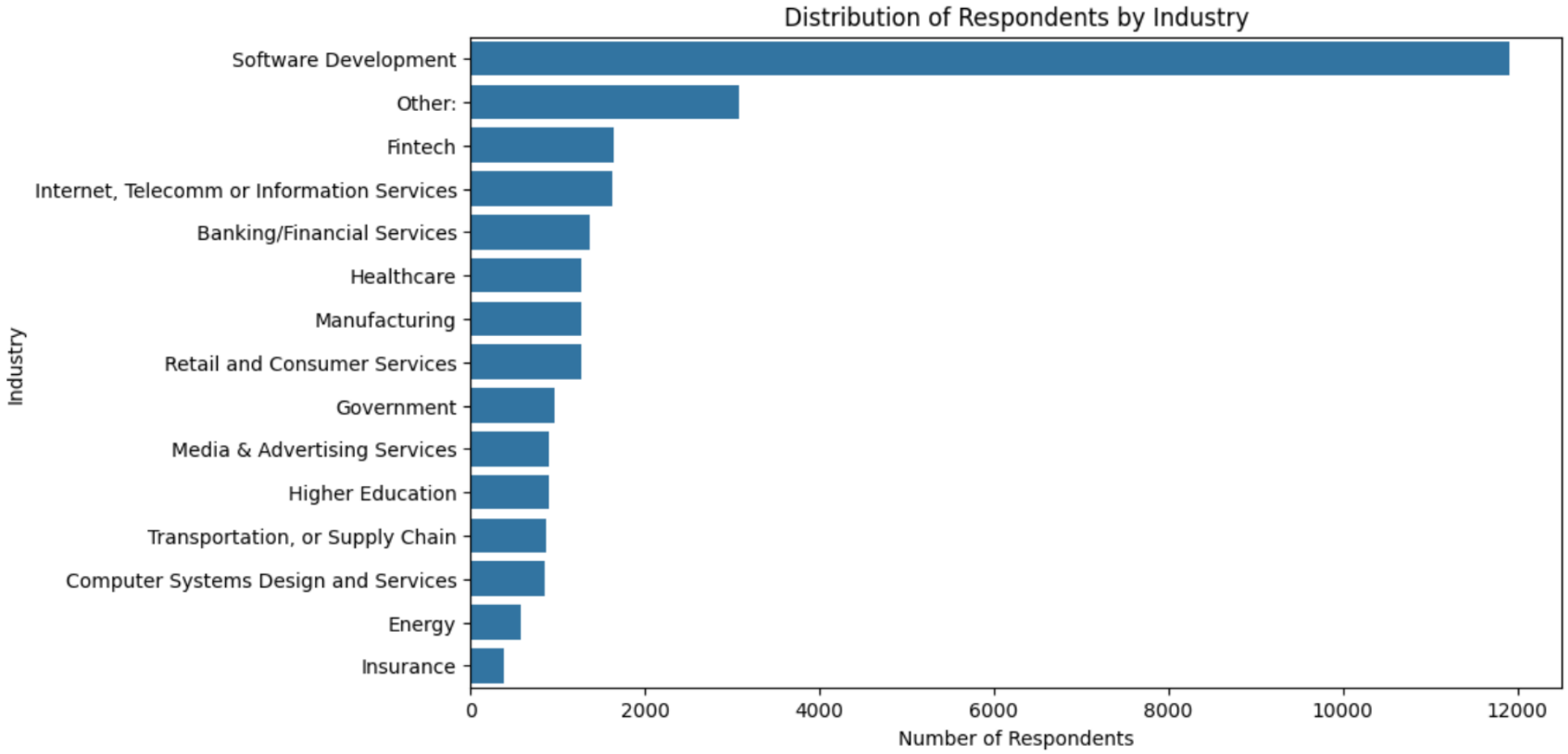

Box Plot of Age (Represented by Midpoints)
20
30
40
50
60
70
Age (Years)

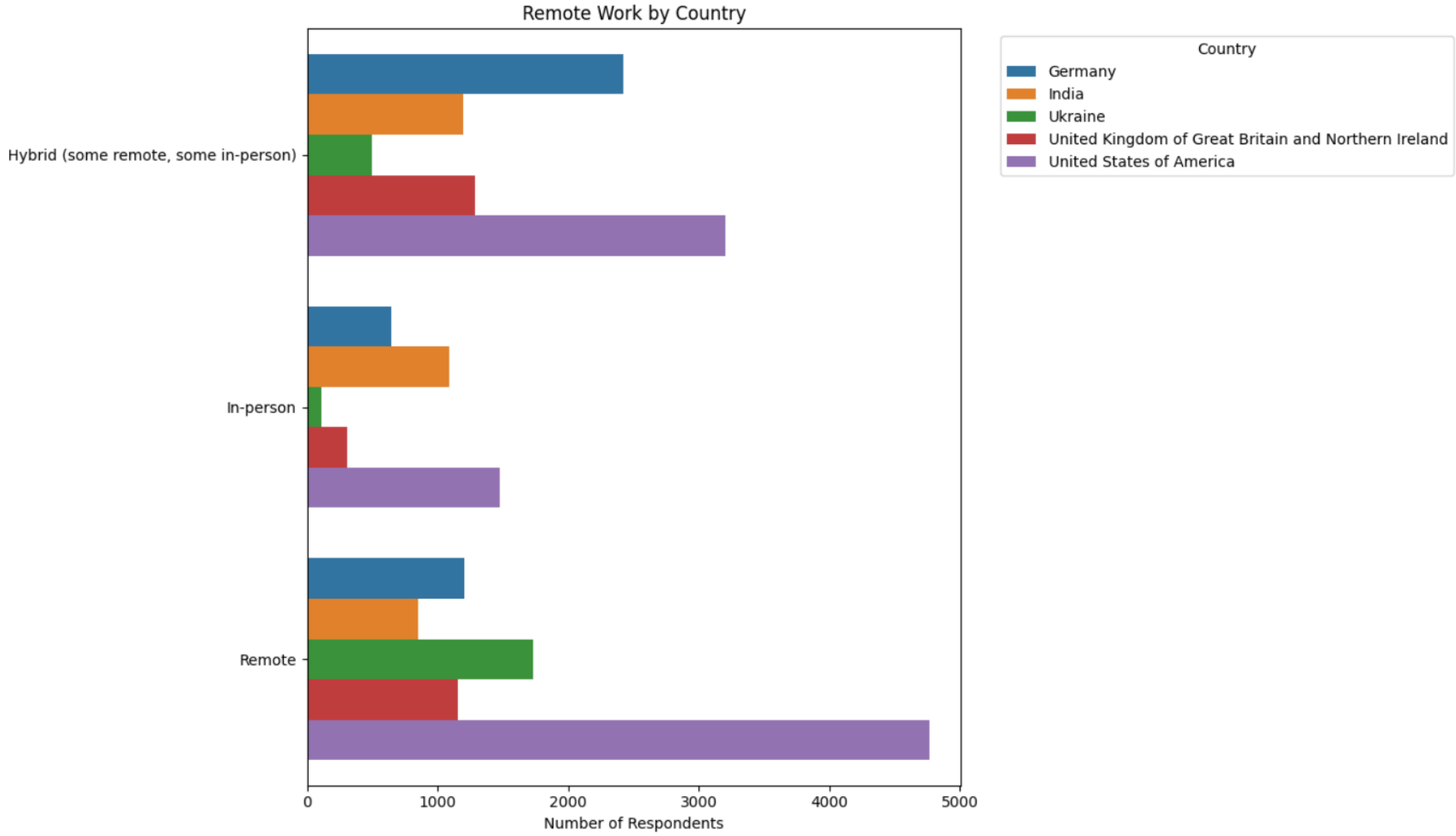

Histogram of Years of Coding Experience
Frequency
14000
12000
10000
8000
6000
4000
2000
0
0
10
20
30
40
50

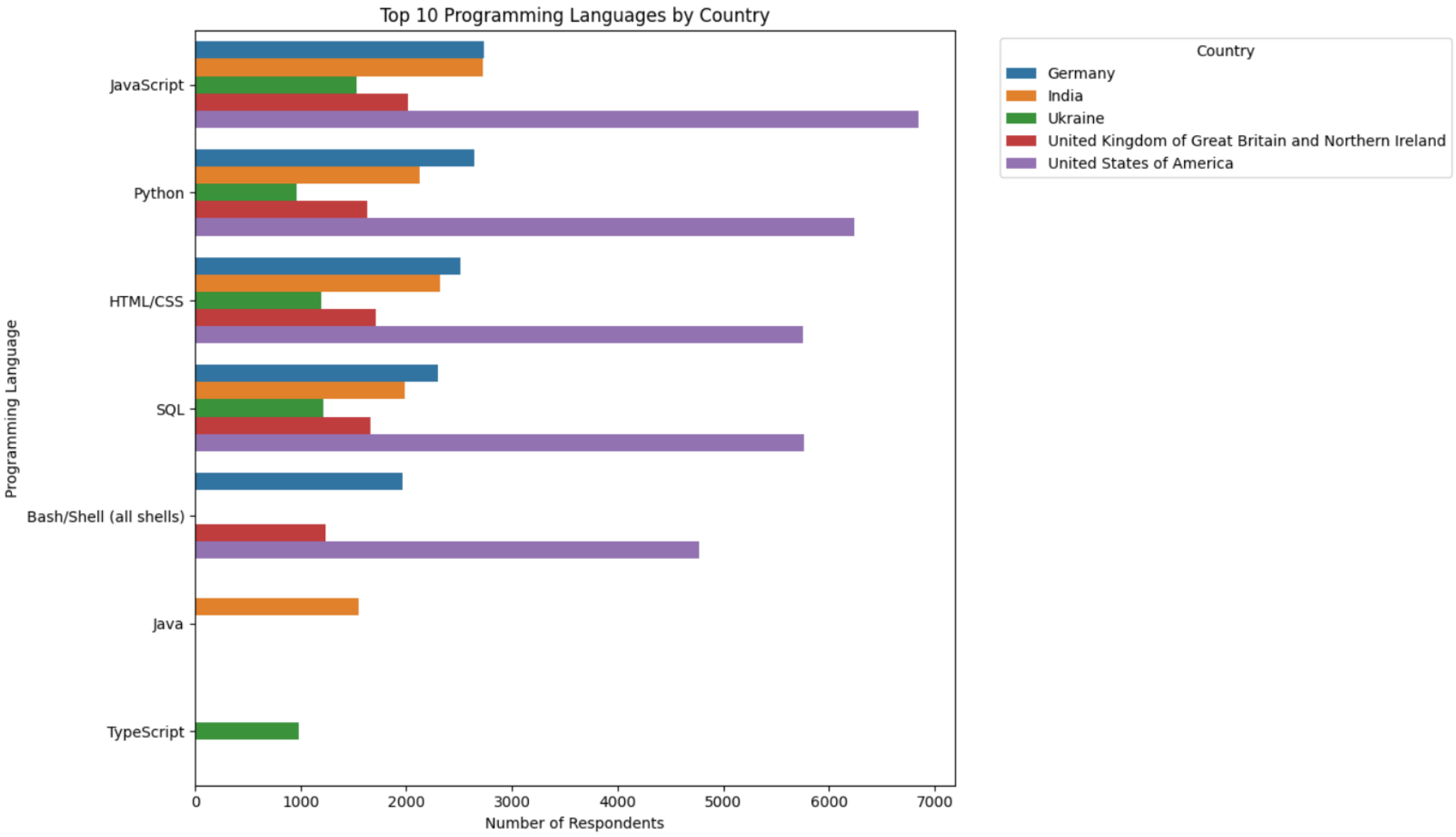

Distribution of Job Satisfaction Levels
Number of Respondents
0
1000
2000
3000
4000
5000
6000
7000
0.0
1.0
2.0
3.0
4.0
5.0
6.0
7.0
8.0
9.0
10.0
Job Satisfaction Score

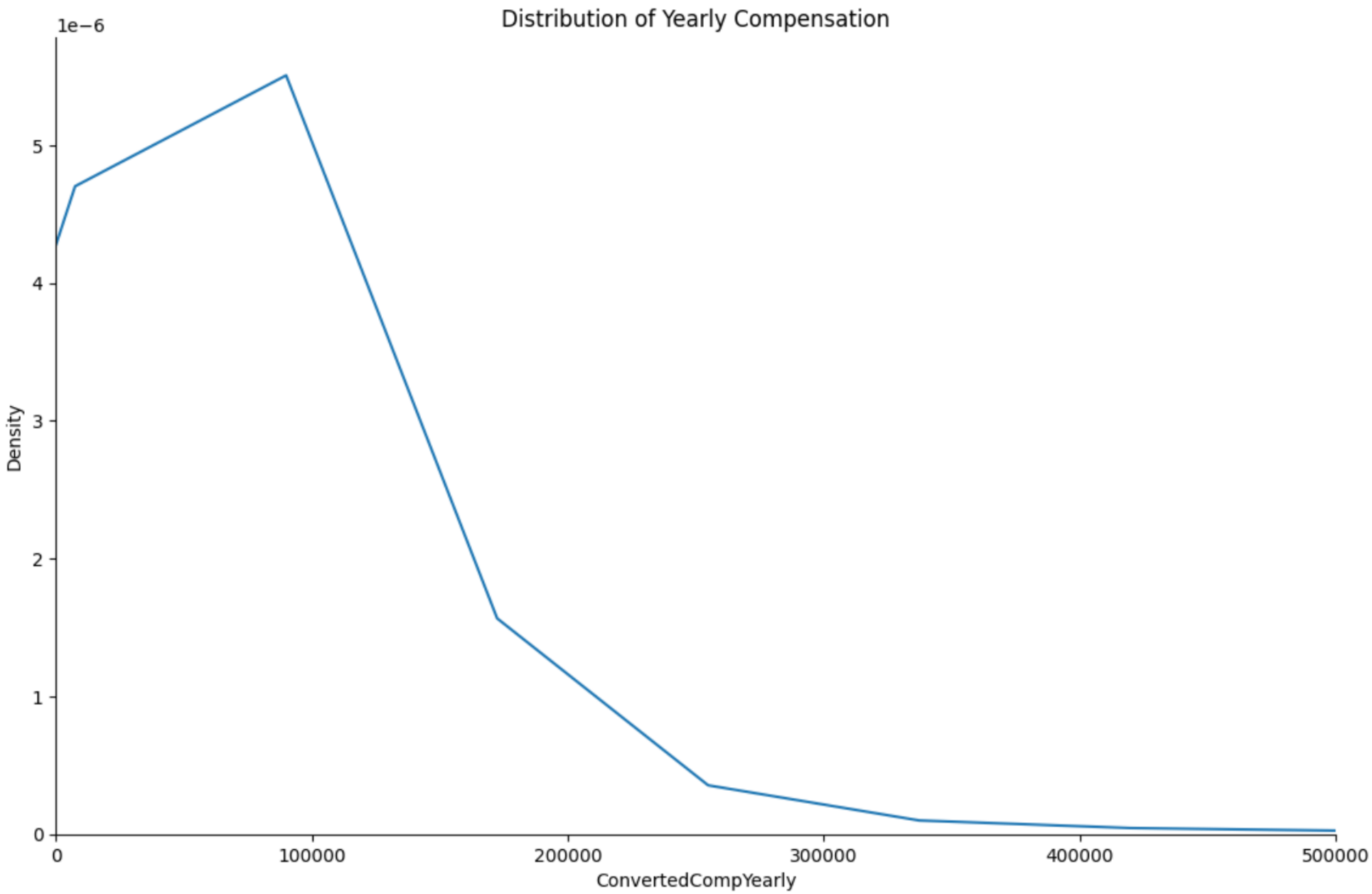

Job Satisfaction Distribution by Professional Coding Experience
JobSat
0.0
1.0
2.0
3.0
4.0
5.0
6.0
7.0
8.0
9.0
10.0
Count of Respondents
0
500
1000
1500
2000
0
10
20
30
40
50
Years of Professional Coding Experience

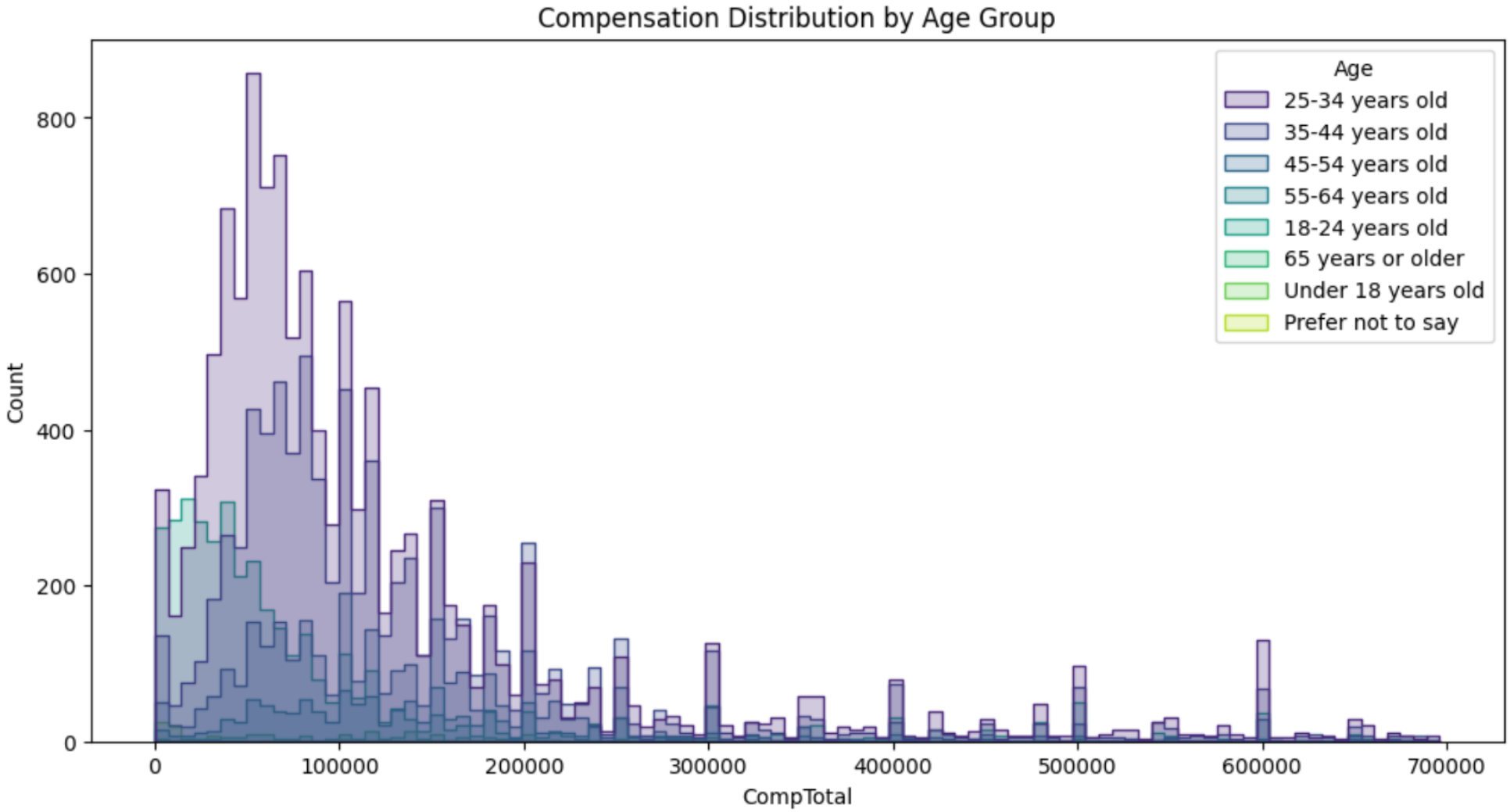

Distribution of Remote Work Preferences
Number of Respondents
0
5000
10000
15000
20000
Remote
In-person
Hybrid
Work Type

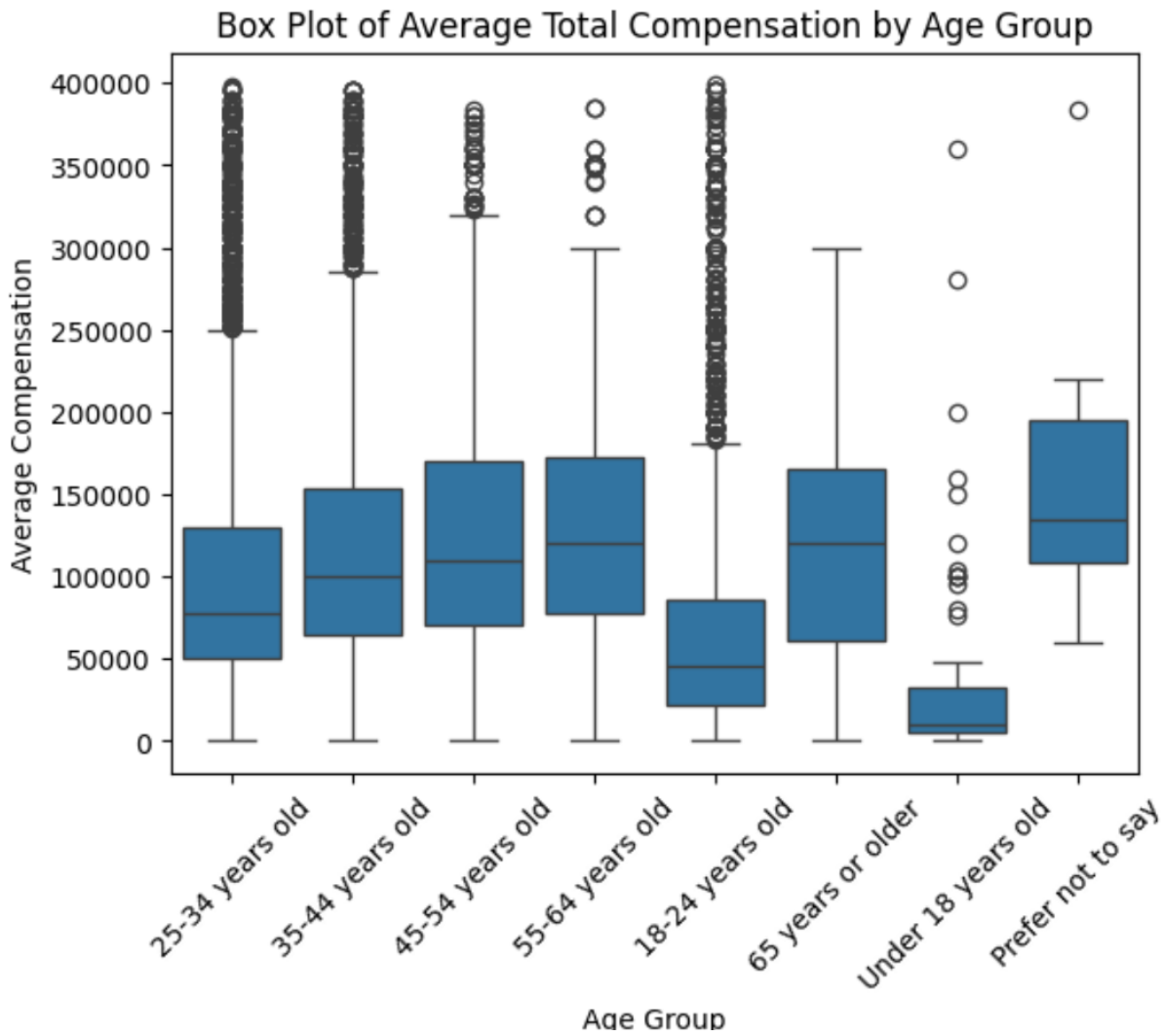

Distribution of Respondents by Industry
Software Development
Other:
Fintech
Internet, Telecomm or Information Services
Banking/Financial Services
Healthcare
Manufacturing
Retail and Consumer Services
Government
Media & Advertising Services
Higher Education
Transportation, or Supply Chain
Computer Systems Design and Services
Energy
Insurance
Industry
0
2000
4000
6000
8000
10000
12000
Number of Respondents

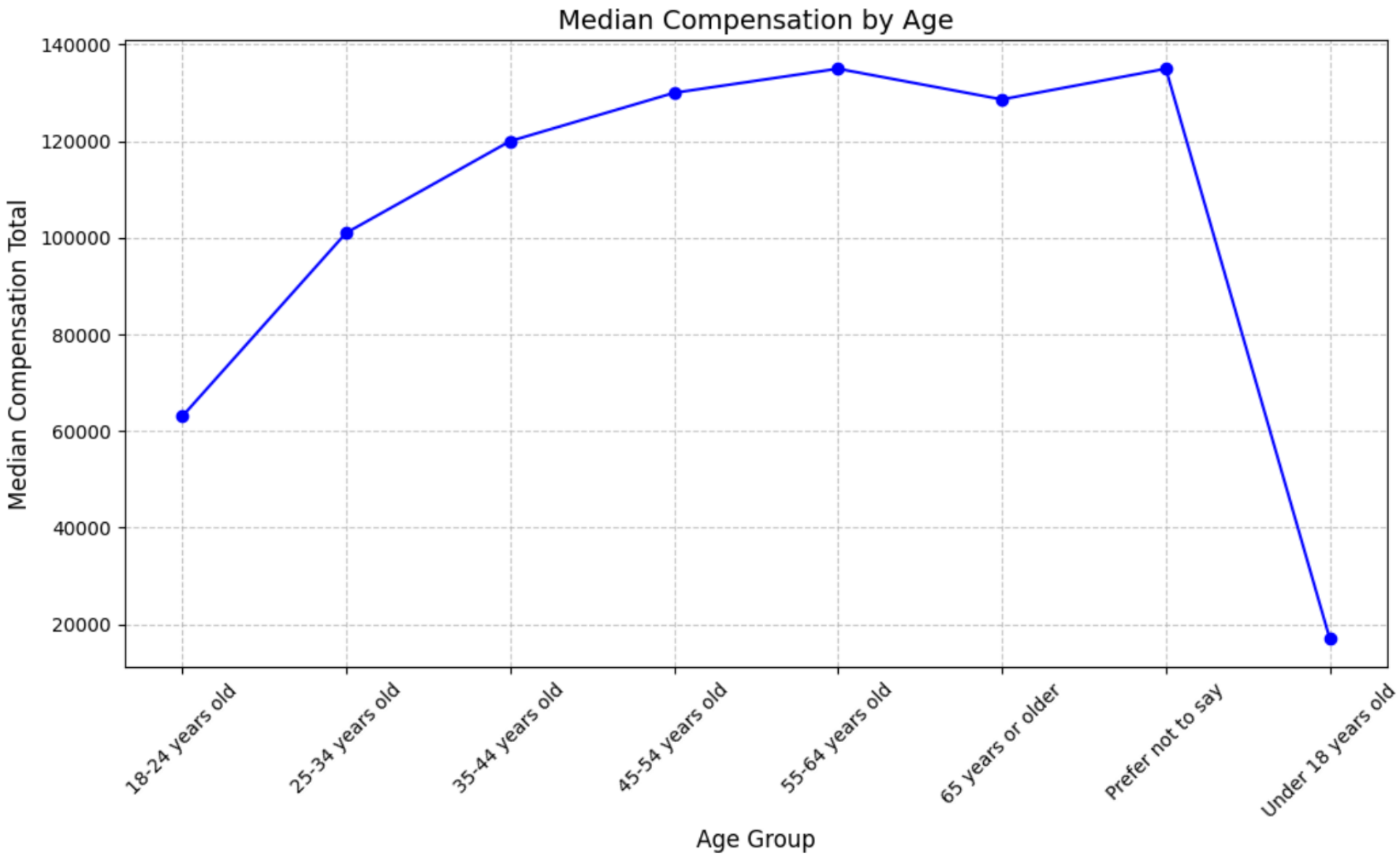

Remote Work by Country
Hybrid (some remote, some in-person)
In-person
Remote
0
1000
2000
3000
4000
5000
Number of Respondents
Country
Germany
India
Ukraine
United Kingdom of Great Britain and Northern Ireland
United States of America

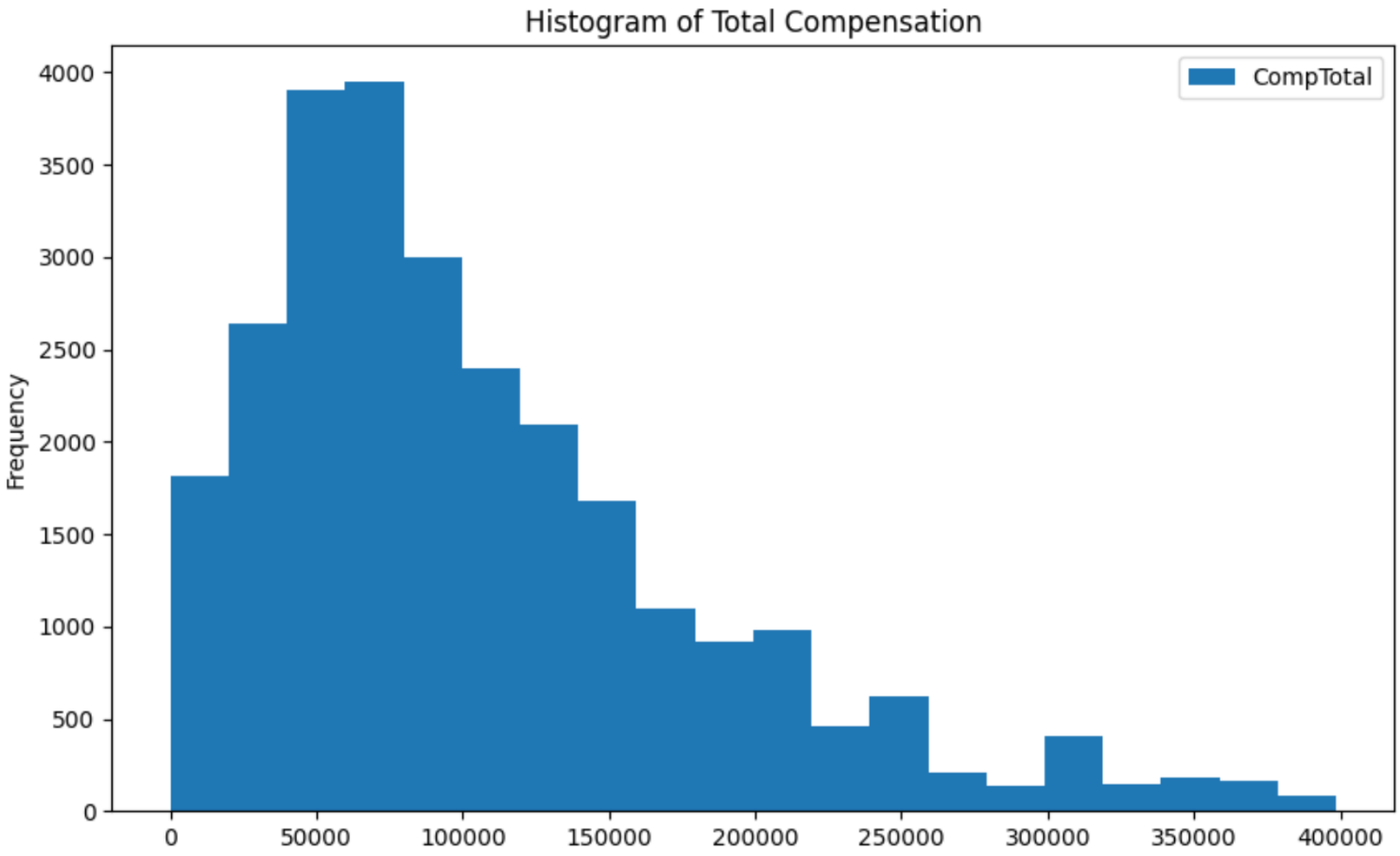

Top 10 Programming Languages by Country
Programming Language
JavaScript
Python
HTML/CSS
SQL
Bash/Shell (all shells)
Java
TypeScript
0
1000
2000
3000
4000
5000
6000
7000
Number of Respondents
Country
Germany
India
Ukraine
United Kingdom of Great Britain and Northern Ireland
United States of America

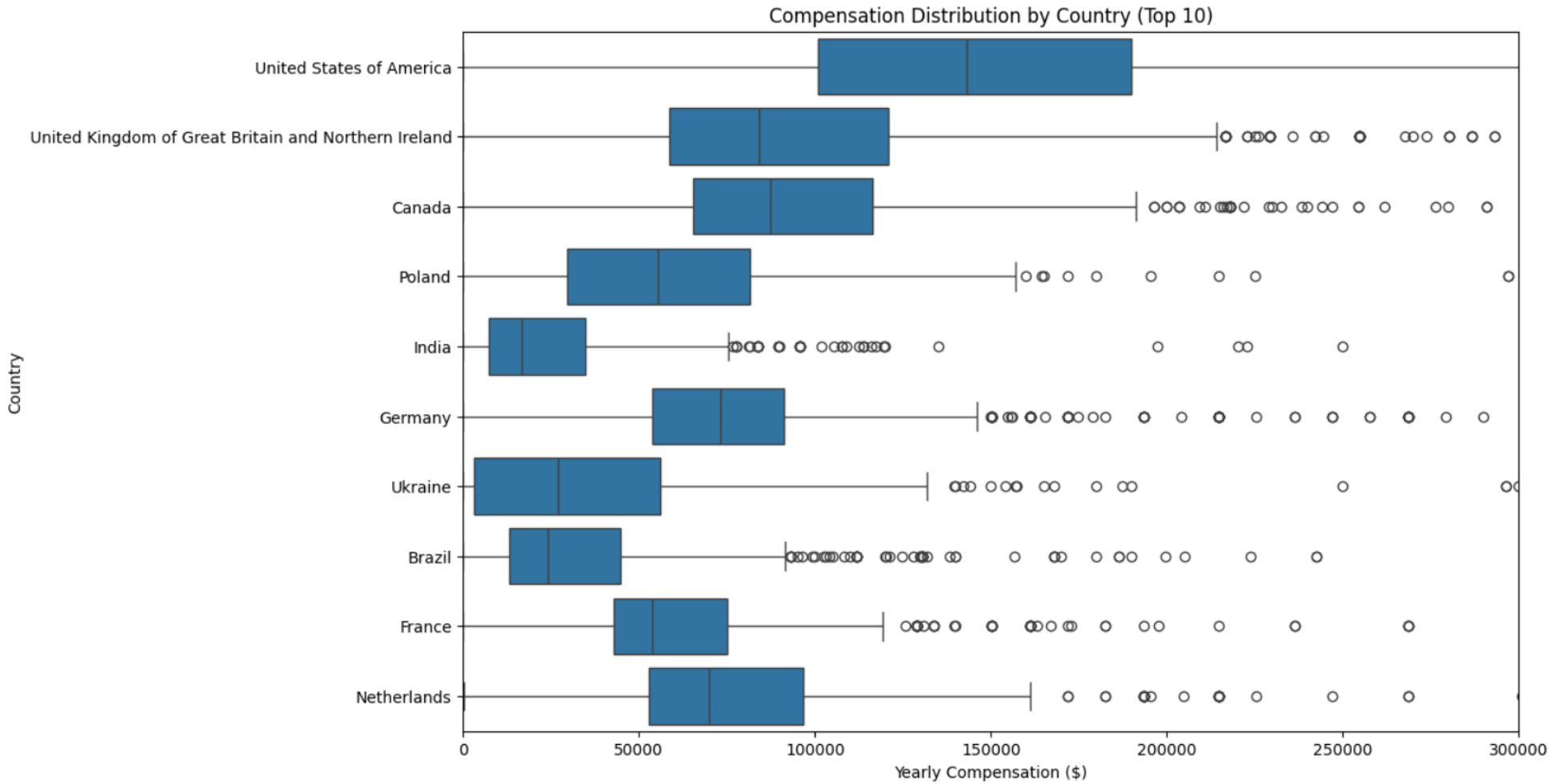

Distribution of Yearly Compensation
1e−6
5
4
3
2
1
0
Density
0
100000
200000
300000
400000
500000
ConvertedCompYearly

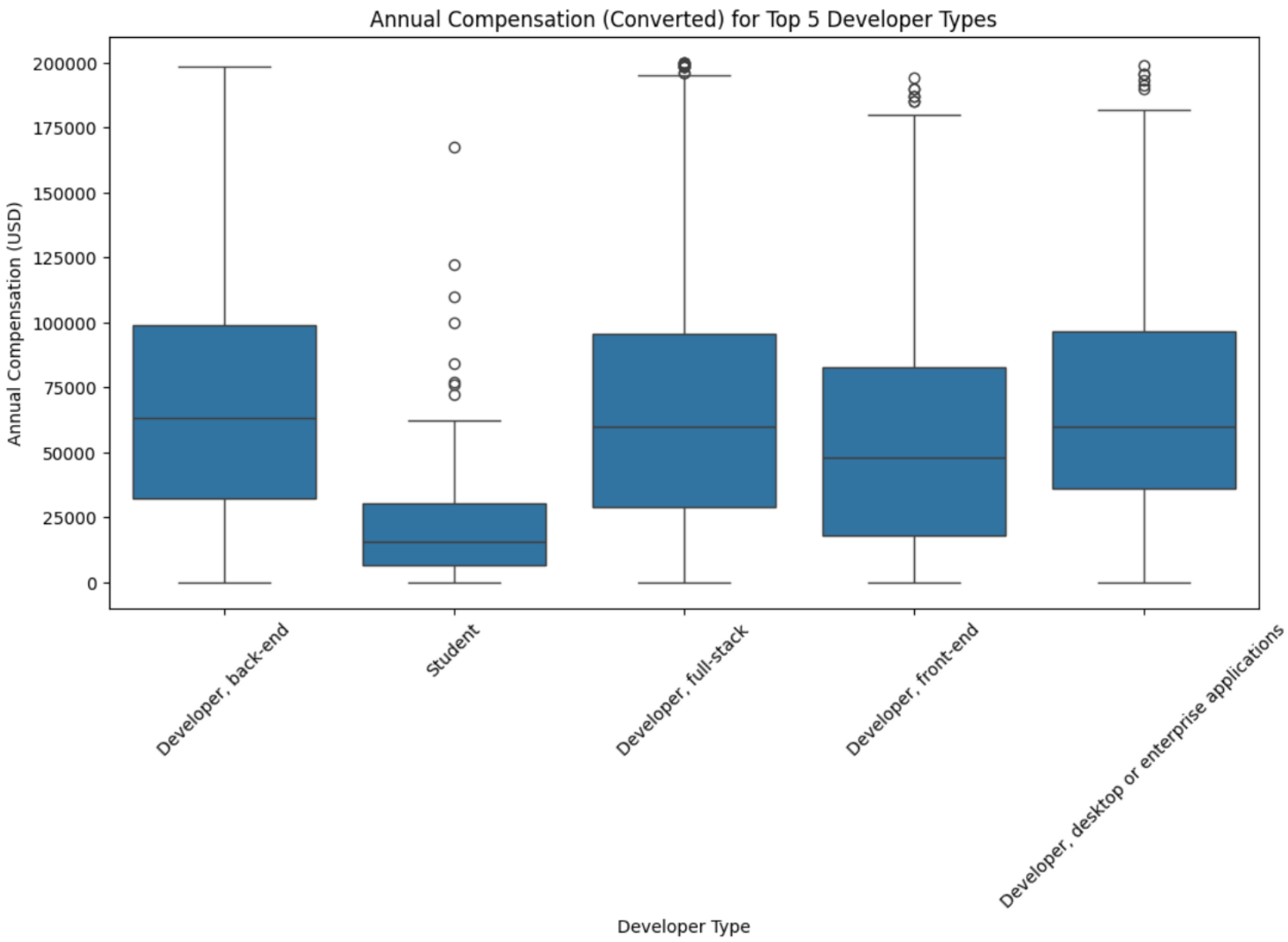

Compensation Distribution by Age Group
Age
25-34 years old
35-44 years old
45-54 years old
55-64 years old
18-24 years old
65 years or older
Under 18 years old
Prefer not to say
Count
0
200
400
600
800
0
100000
200000
300000
400000
500000
600000
700000
CompTotal

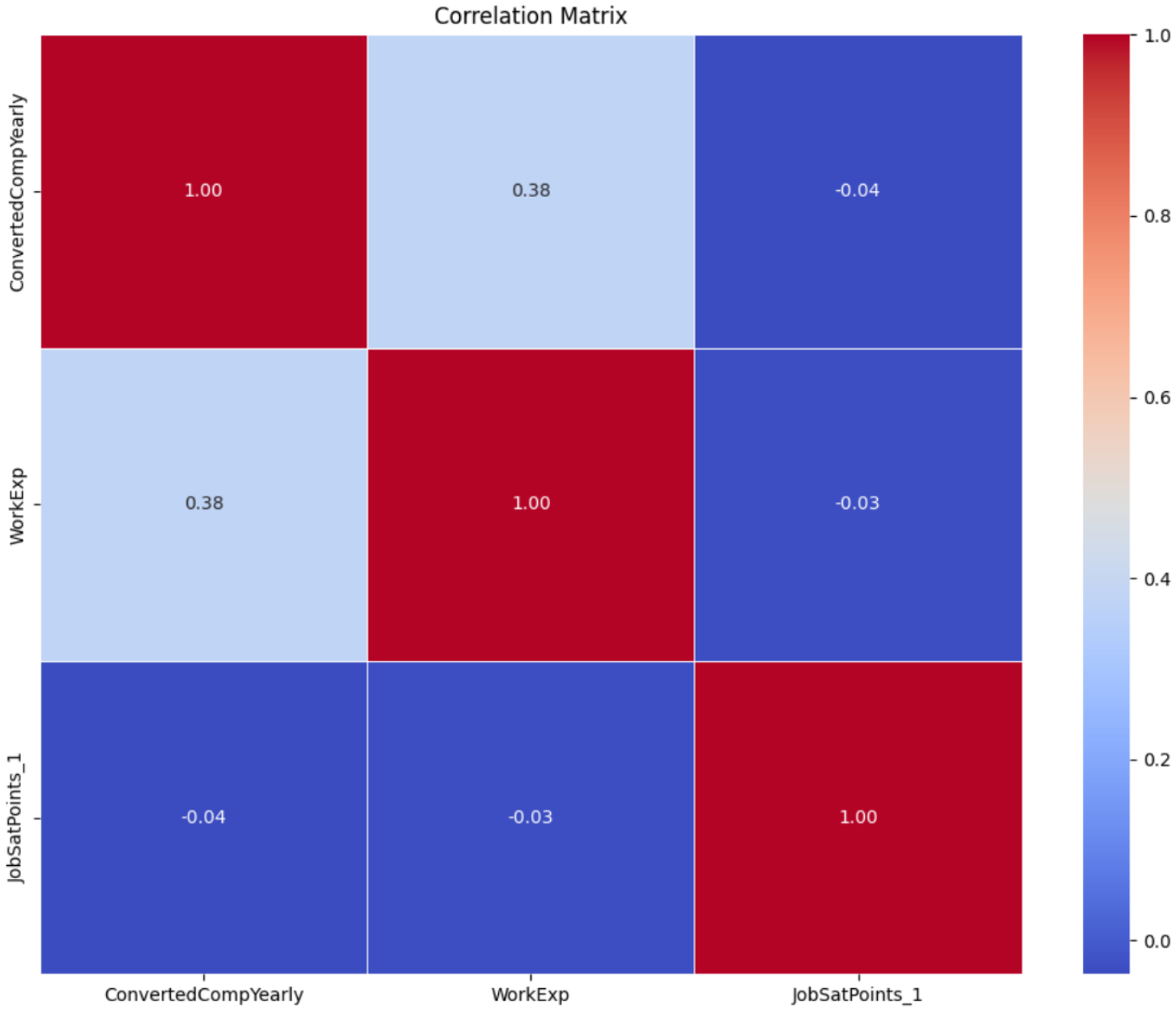

Box Plot of Average Total Compensation by Age Group
Average Compensation
400000
350000
300000
250000
200000
150000
100000
50000
0
25-34 years old
35-44 years old
45-54 years old
55-64 years old
18-24 years old
65 years or older
Under 18 years old
Prefer not to say
Age Group


Median Compensation by Age
Median Compensation Total
140000
120000
100000
80000
60000
40000
20000
18-24 years old
25-34 years old
35-44 years old
45-54 years old
55-64 years old
65 years or older
Prefer not to say
Under 18 years old
Age Group


Histogram of Total Compensation
CompTotal
Frequency
4000
3500
3000
2500
2000
1500
1000
500
0
0
50000
100000
150000
200000
250000
300000
350000
400000


Compensation Distribution by Country (Top 10)
United States of America
United Kingdom of Great Britain and Northern Ireland
Canada
Poland
India
Germany
Ukraine
Brazil
France
Netherlands
Country
0
50000
100000
150000
200000
250000
300000
Yearly Compensation ($)


Annual Compensation (Converted) for Top 5 Developer Types
Annual Compensation (USD)
200000
175000
150000
125000
100000
75000
50000
25000
0
Developer, back-end
Student
Developer, full-stack
Developer, front-end
Developer, desktop or enterprise applications
Developer Type


Correlation Matrix
ConvertedCompYearly
WorkExp
JobSatPoints_1
1.00
0.38
-0.04
0.38
1.00
-0.03
-0.04
-0.03
1.00
ConvertedCompYearly
WorkExp
JobSatPoints_1
1.0
0.8
0.6
0.4
0.2
0.0